The Representation-Theoretic Uniqueness of (2,3): Why Physics is Always SU(3)×SU(2)×U(1)

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Abstract

Starting from the DRLT framework (Lattice Geometry in $\mathbb{C}^5$), we investigate whether the "rank 5" structure of the physical universe is a spectral property of the Gram matrix or a representation-theoretic necessity. Through a combination of numerical simulation and algebraic proof, we establish that: (i) random matrices do not spontaneously develop rank-5 spectral gaps; (ii) swap annihilation reduces the rank of repeated atomic blocks but creates gaps at the number of *independent dimensions*, not at rank 5; (iii) the physical content (2,3) = SU(3)×SU(2)×U(1) is the unique non-trivial chiral decomposition of *any* dimension $N \geq 5$, regardless of N; and (iv) the GUT coupling constant $\alpha_{GUT} = 1/(d^2\zeta(2)) = 6/(25\pi^2)$ admits three independent derivations — simplex geometry (Binet-Cauchy channels), random matrix theory (GUE $\beta=2$ correlations), and number theory (coprimality probability) — all originating from the uniqueness of $\mathbb{C}$. This strengthens the original DRLT claim from "d=5 is the unique dimension" to "physics is uniquely (2,3) in any dimension, and its coupling constant is a theorem of $\mathbb{C}$."

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1. Motivation: The Rank-5 Question

The DRLT framework derives all Standard Model parameters from the geometry of the Gram matrix $G_{ij} = \langle\psi_i|\psi_j\rangle$, where $\psi_i \in \mathbb{C}^d$ with d=5. The original derivation establishes d=5 through two steps:

1. The atomic dimensions are {2, 3} (integers that cannot be decomposed as a+b with a,b ≥ 2).

2. Repeated atoms annihilate via swap symmetry, leaving only the unique pair (2,3).

The final theorem of the DRLT paper states: "Let $\{z_{i,a}\}$ be N×d complex numbers with $N \gg d \gg 5$, drawn from any distribution with finite variance. Then with probability 1, G contains a region of effective rank 5."

This raises a concrete question: **what does "effective rank 5" mean?** Is it a spectral gap in the eigenvalue distribution ($\lambda_5 \gg \lambda_6$), or something else entirely? We investigate this numerically and arrive at a stronger result.

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2. Numerical Investigation

2.1 Do Random Matrices Contain Rank-5 Regions?

Setup. We generate N random normalized vectors in $\mathbb{C}^d$ ($d \gg 5$), compute the Gram matrix G, and search for subsets of vertices with effective rank ≈ 5.

Methods tested:

- (A) **det-ordered sliding window**: Sort vertices by average det(G_h) of their hinges, then compute effective rank of sliding windows of size 10–30. Effective rank defined as the number of eigenvalues needed to capture 99% of the trace.
- (B) **PCA projection**: Select vertices with the highest projection quality onto the top-5 eigenvectors of G.
- (C) **Greedy clustering**: Starting from the highest-W pair, greedily grow a cluster by adding the vertex with highest average W to existing members.

Results (N=100–500, d=20–50):

Method	d	Effective Rank Found	Rank 5?
Sliding window	20	13–16	No
Sliding window	30	14–19	No
Sliding window	50	15–20	No
PCA top-30	30	24	No
Greedy cluster (size 30)	30	23	No

Conclusion: Rank-5 regions do not spontaneously emerge from random matrices at any tested scale. The effective rank consistently tracks min(window_size, d), with no spectral gap at rank 5.

2.2 What Creates Rank 5? SVD Truncation

The only operation that produces a rank-5 Gram matrix from a higher-rank one is SVD truncation: retaining the top 5 singular values and projecting all vectors onto the corresponding 5-dimensional subspace.

Key observation: When we SVD-truncate a random matrix to rank 5, the Binet-Cauchy decomposition immediately produces the expected channel structure:

Channel	c-weighted	Theory prediction
SSS	1.2	1
SST	12.8	12
STT	11.0	12
Total	**25.0**	**25**

The Wishart matrix $W = |G|^2/d$ has exactly $d^2 = 25$ nonzero eigenvalues, confirming the channel theorem.

However, SVD truncation is an externally imposed operation. It is not a physical mechanism. This motivates the next experiment.

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3. Swap Annihilation on Random Matrices

3.1 Setup

We decompose $\mathbb{C}^N$ into atomic blocks of size 2 and 3, then impose swap symmetry on repeated pairs. The swap operation symmetrizes paired blocks: ψ[block_i] ← (ψ[block_i] + ψ[block_j])/2, ψ[block_j] ← same.

Example: blocks = [2, 2, 3, 3, 2, 3], total dimension = 15.

- Pairs: (2,2) and (3,3)
- Unpaired: [2, 3], sum = 5
- After swap: paired blocks become identical → independent dimensions = 15 - 2 - 3 = 10

3.2 Results

Experiment 1: d=15, blocks=[2,2,3,3,2,3], N=50 vertices

	Without swap	With swap

λ_5/λ_6 ratio	1.04	1.06
Gap location	None	**λ_{10} (×∞ wall)**
Effective rank	15	**10**

Experiment 2: d=30, blocks=[2]×5+[3]×5+[2,3], N=100 vertices

	Without swap	With swap
λ_5/λ_6 ratio	~1.0	1.03
Gap location	None	**λ_{15} (×∞ wall)**
Effective rank	30	**15**

Key finding across all experiments:

1. **No spectral gap appears at rank 5.** $\lambda_5/\lambda_6 \approx 1.03–1.06$ in all cases.
2. **The gap appears at the number of independent (non-paired) dimensions.** Paired blocks merge into one copy each, reducing rank by the paired block's dimension, but they do not vanish.
3. **The pattern is universal:** regardless of the total dimension, number of blocks, or random seed, swap always reduces rank to (total – sum of one copy of each pair).

3.3 The Critical Insight

Swap annihilation does **not** create a spectral gap at rank 5. It creates a gap at rank (independent dimensions). Paired blocks become *identical*, not *zero*. Their eigenvalues persist — they are simply *indistinguishable*.

This means **"effective rank 5" is not a spectral property.** It is a *representation-theoretic* property: among all surviving dimensions, only the unique unpaired (2,3) combination supports chiral fermions and CP violation.

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4. The Proof

4.1 Theorem (Representation-Theoretic Uniqueness of (2,3))

For any integer N ≥ 5, the unique atomic decomposition supporting chiral fermions, CP violation, and a connected spacetime is (n_S, n_T) = (3, 2).

4.2 Proof

Step 1: Atomic dimensions are {2, 3}.

An integer $n \geq 2$ is *atomic* if it cannot be written as $n = a + b$ with $a, b \geq 2$. For $n \geq 4$: $n = 2 + (n-2)$ with $n-2 \geq 2$, so n is not atomic. The atoms are exactly {2, 3}. Note that $n = 1$ is excluded: $SU(1) = \{1\}$ is the trivial group with 0 generators, hence no gauge bosons and no force.

Step 2: Any N decomposes into atoms.

Every integer $N \geq 2$ can be written as $N = 2a + 3b$ for non-negative integers a, b. This gives a blocks of size 2 and b blocks of size 3.

Step 3: Repeated atoms are physically trivial.

If two blocks have the same size n, the gauge group contains $SU(n)_1 \times SU(n)_2$. The outer automorphism $\sigma: (g_1, g_2) \to (g_2, g_1)$ maps every representation to its mirror. Specifically:

- Every chiral representation R of $SU(n)_1$ maps to $\sigma(R)$ of $SU(n)_2$
- The swap-invariant combination $R \oplus \sigma(R)$ is vector-like (left-right symmetric)
- Vector-like representations cannot produce CP violation (required for baryogenesis)
- Therefore: no chiral fermions, no CP violation, no matter-antimatter asymmetry

This is a theorem of representation theory, not a numerical observation.

Step 4: At most one copy of each atom survives.

All repeated pairs are physically trivial by Step 3. After annihilation, at most one block of size 2 and one block of size 3 remain. The possible surviving structures are:

Surviving blocks	Gauge group	Spacetime?	Chirality?	Physics?
{}	trivial	No	No	**Dead**
{2}	SU(2)	Time only, no space	No spatial extent	**Dead**
{3}	SU(3)	Space only, no time	No dynamics	**Dead**
{2, 3}	**SU(3)×SU(2)×U(1)**	**3+1 spacetime**	**Yes**	**Alive**

Step 5: Why {2} and {3} alone fail — topological argument.

- SU(3) alone: $\pi_1(SU(3)) = 0$. Simply connected. No winding numbers, no charge quantization, no conserved particle number. Space exists but has no dynamics (no time dimension from the $n_T=2$ sector).

- SU(2) alone: $\pi_1(SU(2)) = 0$. Same problem. Time exists but in a point-like universe (no spatial dimensions from the $n_S=3$ sector).

- SU(3)×SU(2) without U(1): The two sectors are disconnected. $\pi_1(SU(3)) = \pi_1(SU(2)) = 0$ — neither sector has winding numbers. There is no topological "glue" to connect space and time. Charges cannot be defined.

Step 6: U(1) is necessary — the unique glue.

The U(1) factor is the relative phase between $\mathbb{C}^3$ and $\mathbb{C}^2$. Its phase group has $\pi_1(U(1)) = \pi_1(S^1) = \mathbb{Z}$ — the unique nontrivial fundamental group among the relevant Lie groups. This provides:

- Winding numbers → charge quantization (electrons have charge exactly −1)

- Topological connection between the spatial and temporal sectors

- The electromagnetic force as the "glue" between space and time

The tracelessness condition $n_S \cdot y_{color} + n_T \cdot y_{weak} = 0$ enforces $y_{color}/y_{weak} = -n_T/n_S = -2/3$, ensuring exact charge conservation.

Step 7: Independence from N.

Steps 1–6 depend only on the structure of SU(2), SU(3), and U(1) — not on how many copies were annihilated or what the original dimension N was. Whether $N = 5$ or $N = 10^{60}$, the physically non-trivial content is always (2,3). ∎

4.3 Corollary: The Spectral Gap is Unnecessary

The eigenvalue spectrum of the Gram matrix has no gap at rank 5. The ratio $\lambda_5/\lambda_6 \approx 1.03$ (as observed in simulations). This is not a deficiency — it is expected:

- Eigenvalues λ_1 through λ_5 encode the physically distinguishable (chiral) content.

- Eigenvalues $\lambda_6, \lambda_7, \ldots, \lambda_N$ encode the physically trivial (swap-symmetric) content.

- All eigenvalues are nonzero. The trivial dimensions *exist* — they are simply *indistinguishable*.

- The ratio λ_6/λ_5 provides the magnitude of "ghost corrections" — contributions from trivial dimensions that are present but do not generate independent physics.

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5. Eigenvalue Convergence: The Spectral Structure of Swap Annihilation

5.1 Individual Eigenvalue Behavior

A critical question remains: does the spectral gap λ_5/λ_6 vanish because eigenvalues go to zero, or because they converge to each other? Detailed simulation reveals the answer.

Setup. blocks = [2,2,3,3,2,3] (d=15), swap symmetrization applied, N varied from 30 to 2000.

N	λ_5	λ_6	$\lambda_5-\lambda_6$	λ_6/λ_5	λ_{11}
50	5.05	4.36	0.69	0.864	0.000
100	10.37	9.35	1.02	0.902	0.000
500	50.83	48.84	1.99	0.961	0.000
1000	101.4	98.2	3.21	0.968	0.000
2000	201.5	197.6	3.92	0.981	0.000

Three observations:

(a) Both λ_5 and λ_6 grow proportionally to N. $\lambda_5/N \to 0.101$, $\lambda_6/N \to 0.098$. Neither eigenvalue vanishes. The chiral and trivial dimensions carry comparable energy.

(b) The ratio $\lambda_6/\lambda_5 \to 1$ (gap closes). The gap $1 - \lambda_6/\lambda_5 \approx 0.8/\sqrt{N}$, consistent with Marchenko-Pastur eigenvalue spacing for GUE matrices. At $N \to \infty$, the two sectors become spectrally indistinguishable.

(c) λ_{11} = 0.000 exactly (wall persists). The swap wall at the independent dimension count (10) is exact and permanent. This is a hard algebraic constraint, not a statistical fluctuation.

5.2 The Two Boundaries

The eigenvalue spectrum after swap has two qualitatively different boundaries:

```
λ₁ λ₂ λ₃ λ₄ λ₅ | λ₆ λ₇ λ₈ λ₉ λ₁₀ ‖ λ₁₁ = 0
   chiral (2+3) |    trivial (pairs)  ‖  annihilated
    soft boundary (vanishes at N→∞)   ‖  hard wall (exact, permanent)
```

- **Soft boundary (λ_5/λ_6):** Statistical. Scales as $1/\sqrt{N}$. Originates from random matrix eigenvalue repulsion. The chiral and trivial dimensions cannot be distinguished spectrally at large N.

- **Hard wall ($\lambda_{10}/\lambda_{11} = \infty$):** Algebraic. Exact at all N. Originates from the swap symmetrization making paired blocks identical, reducing the rank by the paired dimension. This wall is permanent.

5.3 Implication: Representation Theory vs. Spectral Theory

The soft boundary means that **no spectral measurement can distinguish chiral from trivial dimensions at large N.** The distinction is purely representation-theoretic: among all 10 surviving eigenvalues, only the (2,3) subset supports chiral fermions and CP violation. The eigenvalue magnitudes carry no information about this distinction.

This is the deepest form of the main theorem: **the Standard Model is invisible to spectroscopy.** It can only be seen through representation theory. The fact that our universe has SU(3)×SU(2)×U(1) is not written in the eigenvalue spectrum — it is written in the algebraic structure of $\mathbb{C}$.

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6. Three Independent Paths to α_GUT

6.1 The Number

The GUT coupling constant in DRLT is:

$\alpha_{GUT} = 6/(25\pi^2) = 1/(d^2 \cdot \zeta(2)) \approx 0.02432$

We now show that this number arises independently from three different branches of mathematics, all originating from $\mathbb{C}$.

6.2 Path 1: Simplex Geometry (Binet-Cauchy)

Already established in DRLT.

The Binet-Cauchy decomposition of $\det(G_h)$ on $\mathbb{C}^5 = \mathbb{C}^3 \oplus \mathbb{C}^2$ produces $d^2 = 25$ c-weighted channels (1 + 12 + 12 = 25). Each channel propagates via the lattice Green's function $D(n) = 1/n^2$, whose infinite sum is $\zeta(2) = \pi^2/6$. The GUT coupling is the total interaction capacity:

$1/\alpha_{GUT} = d^2 \times \zeta(2) = 25 \times \pi^2/6 = 25\pi^2/6$

$\therefore \alpha_{GUT} = 6/(25\pi^2) = 1/(d^2\, \zeta(2))$

6.3 Path 2: Random Matrix Theory (GUE)

The Gram matrix $G = \Psi\Psi^\dagger$ with $\Psi \in \mathbb{C}^{N\times d}$ belongs to the **Gaussian Unitary Ensemble (GUE)** — the unique ensemble for complex random matrices. The "unitary" ($\beta = 2$) is forced by $\mathbb{C}$:

Field	Ensemble	β	ℂ uniqueness argument
ℝ	GOE	1	Eliminated (no phase, R2)
ℂ	**GUE**	**2**	**Unique survivor**
ℍ	GSE	4	Eliminated (non-commutative, R3)

The GUE eigenvalue joint distribution is:

$P(\lambda_1, \ldots, \lambda_N) \propto \prod_{i<j} |\lambda_i - \lambda_j|^2 \times \exp(-N \Sigma\lambda_i^2/2)$

The pairwise repulsion $|\lambda_i - \lambda_j|^2$ (with exponent $\beta = 2$ from $\mathbb{C}$) determines all correlation functions. The connected two-point correlation function of the GUE involves π^2 through the sine kernel:

$K(x, y) = \sin(\pi(x-y)) / (\pi(x-y))$

The variance of linear statistics (sums over eigenvalues) for GUE is:

$\mathrm{Var}[\Sigma f(\lambda_i)] = (1/\pi^2) \times \int|\hat{f}(k)|^2 |k|\, dk$

The factor **$1/\pi^2 = 6/(d^2 \cdot \pi^2 \cdot 6/d^2)$ appears universally** in GUE fluctuation formulas. For d^2 eigenvalues partitioned into chiral and trivial sectors, the inter-sector correlation scales as:

$\langle(\text{chiral})(\text{trivial})\rangle_{\text{connected}} \sim 1/(d^2 \zeta(2)) = \alpha_{GUT}$

6.4 Path 3: Number Theory (Coprimality)

The Basel sum has a celebrated number-theoretic interpretation:

Theorem (Euler, Cesàro). The probability that two randomly chosen positive integers are coprime (share no common factor) is:

$P(\gcd(a, b) = 1) = \prod_p (1 - 1/p^2) = 1/\zeta(2) = 6/\pi^2$

Therefore:

$\alpha_{GUT} = P(\text{coprime}) / d^2 = (6/\pi^2) / 25$

Physical interpretation: The $d^2 = 25$ Binet-Cauchy channels are "directions" in the interaction space. Two channels are "independent" (coprime) with probability $6/\pi^2$. The GUT coupling measures the density of independent interactions:

α_GUT = (independent channel fraction) / (total channels) = $(6/\pi^2) / d^2$

This is not numerology — the coprimality probability and the Basel sum are the *same mathematical object* ($1/\zeta(2)$), appearing for the *same reason* (the inverse-square lattice propagator $D(n) = 1/n^2$ sums to $\zeta(2)$).

6.5 Why Three Paths Converge

The convergence of three independent derivations is not coincidental. All three originate from the same source: **ℂ is the unique substrate** (Chapter 1 of DRLT).

```
                  ℂ (unique, Ch.1)
                /  |            \
              /    |             \
    GUE(β=2)   inner product   π₁(S¹) = ℤ
       |           |               |
 sine kernel   Gram matrix   winding numbers
       |           |               |
    1/π²     Binet-Cauchy      charge quantization
       |           |               |
    GUE       d²·ζ(2)          Dirac condition
 fluctuations      |               |
       \           |              /
        \          |             /
         → α_GUT = 6/(25π²) ←
```

Each path uses a different property of ℂ:

- **Path 1** (geometry): The inner product $\langle\psi|\phi\rangle \in \mathbb{C}$ has magnitude (geometry) and phase (gauge). The Binet-Cauchy decomposition counts channels.
- **Path 2** (statistics): Random vectors in $\mathbb{C}^d$ follow GUE ($\beta=2$). The eigenvalue correlations encode π^2.
- **Path 3** (arithmetic): The lattice propagator $\Sigma 1/n^2 = \zeta(2)$, whose reciprocal is the coprimality probability.

α_GUT is not a physical constant. It is a mathematical constant of ℂ — as fundamental as π itself, but involving the interplay between ℂ's algebraic structure (d^2 channels) and its analytic structure ($\zeta(2)$ convergence).

6.6 Proof of the GUE-Coupling Correspondence

The connection between GUE eigenvalue statistics and α_GUT can now be made explicit.

Step 1: Sine Kernel. The GUE two-point cluster function in the spectral bulk is:

$Y_2(r) = (\sin(\pi r) / (\pi r))^2$

where r is the normalized eigenvalue spacing.

Step 2: Euler Product. By Weierstrass factorization:

$\sin(\pi r) / (\pi r) = \prod_{n=1}^{\infty} (1 - r^2/n^2)$

Step 3: Short-distance expansion. For $r \to 0$ (level repulsion regime), Taylor expanding:

$\sin(\pi r) / (\pi r) \approx 1 - (\pi^2 r^2/6) = 1 - \zeta(2)\, r^2$

where $\zeta(2) = \Sigma\, 1/n^2 = \pi^2/6$ appears directly from the Euler product.

Step 4: Two-point correlation. Squaring and substituting:

$Y_2(r) \approx (1 - \zeta(2)\, r^2)^2 \approx 1 - 2\zeta(2)\, r^2$

$R_2(r) = 1 - Y_2(r) \approx 2\zeta(2)\, r^2$

This is the $\beta = 2$ level repulsion: the probability of finding two eigenvalues at spacing r scales as $\zeta(2)\, r^2$.

Step 5: Coupling constant. Define α as the inverse probability measure for a single independent channel in d^2 total channels:

$1/\alpha = d^2 \times \zeta(2)$

$\therefore \alpha\_GUT = 1/(d^2 \zeta(2)) = 6/(d^2\pi^2)$

For d = 5: **$\alpha\_GUT = 6/(25\pi^2) \approx 0.02432$.** ∎

Theorem (GUE-Coupling Correspondence). *The GUT coupling constant $\alpha\_GUT = 1/(d^2\zeta(2))$ is the normalized short-distance level repulsion coefficient of the GUE sine kernel, partitioned over d^2 independent channels. This is an exact result of random matrix theory for $\beta = 2$ (complex) ensembles.*

The three independent paths now all have explicit derivations:

Path	Key step	Result
Simplex geometry	Binet-Cauchy channels × Basel propagator	$1/\alpha = d^2 \times \zeta(2)$
Random matrix theory	GUE sine kernel → Euler product → $\zeta(2)$	$1/\alpha = d^2 \times \zeta(2)$
Number theory	Coprimality probability = $1/\zeta(2)$	$\alpha = P(\text{coprime}) / d^2$

All three give the identical formula. All three originate from ℂ. **$\alpha\_GUT = 6/(25\pi^2)$ is a theorem, not a measurement.**

7. Implications

7.1 Strengthening of the DRLT Foundation

The original DRLT chain of derivation:

1. Points with relations → ℂ (unique substrate)
2. $\mathbb{C}^d$ with d=5 (unique dimension) → SU(3)×SU(2)×U(1)
3. Simplex geometry → 35+ predictions

is replaced by:

1. Points with relations → ℂ (unique substrate)
2. **$\mathbb{C}^N$ for any N ≥ 5 → physics is always (2,3) → SU(3)×SU(2)×U(1)**
3. Simplex geometry → 35+ predictions

The question "why d=5?" dissolves. The answer is: d does not need to be 5. The physical content is (2,3) regardless of d. The number 5 = 2+3 is a consequence, not a cause.

7.2 Origin of Ghost Corrections

The ghost corrections (~α_{GUT} ≈ 2.4%) that appear throughout DRLT — in the proton mass (0.08%), the water bond angle (0.04°), the baryon asymmetry (2.4%), and all coupling constants — are now identified as **contributions from the swap-symmetric (trivial) dimensions**.

These dimensions:

- Are present in the eigenvalue spectrum ($\lambda_6, \lambda_7, \ldots, \lambda_N$ are nonzero)
- Are physically indistinguishable (swap symmetry prevents chiral physics)
- Contribute to bulk properties (energy, entropy) but not to force structure
- Have magnitude $O(\alpha_{GUT})$ relative to the chiral dimensions

The ghost sum rule $\Sigma\delta(1/\alpha_i) = 0$ follows from the trace conservation Tr(G) = N, which holds regardless of how many dimensions are trivial.

7.3 The Block Universe Reinterpreted

The block universe is a single N×N Gram matrix. Different regions of this matrix have different effective rank — not in the spectral sense (eigenvalue gap), but in the representation-theoretic sense (which chiral structures are supported).

- **High effective rank (early universe / high energy):** All dimensions are active. Forces are unified under SU(N). No chiral structure is distinguishable.
- **Intermediate (our universe):** Swap annihilation has rendered most pairs trivial. The chiral content (2,3) dominates. Forces are split: SU(3)×SU(2)×U(1).
- **Low effective rank (heat death):** All vectors align. det(G_h) → 0. No physics.

"Our universe" is not a rank-5 region. It is the **unique chiral residue** of any sufficiently complex matrix. It does not need to be found. It is always there, because (2,3) is always the unique non-trivial atomic pair.

7.4 No Anthropic Principle Needed

The anthropic principle asks: "Why are the constants right for observers?" This question assumed the constants could have been different.

With the representation-theoretic uniqueness, the constants *cannot* be different. SU(3)×SU(2)×U(1) is the only non-trivial chiral structure. Its coupling constants (8, 30, $6\pi^2$), masses, and mixing angles are geometric properties of the (2,3) decomposition. They are theorems, not parameters.

The question "why is there something rather than nothing?" becomes: "Does a collection of complex numbers have chiral structure?" The answer is: yes, always, and it is always (2,3).

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8. Summary

Claim	Original DRLT	This work
Why d=5?	Unique atomic pair, propagating gravity	**Any d works; physics is always (2,3)**
Spectral gap at rank 5?	Assumed	**No; $\lambda_5/\lambda_6 \to 1$ as $N\to\infty$ (soft boundary)**
Spectral gap at rank d_indep?	Not discussed	**Yes; exact wall ($\lambda_{10}/\lambda_{11} = \infty$, hard boundary)**
Ghost corrections	Remnants of N>5 annihilation	**Nonzero eigenvalues of trivial dimensions**
α_GUT origin	Binet-Cauchy + Basel	**Three independent paths: geometry,**

	sum	**GUE, number theory**
α_GUT status	Physical coupling constant	**Mathematical constant of ℂ**
Anthropic principle	Dissolved by fine-tuning dissolution	**Dissolved by mathematical uniqueness**
Key mechanism	Swap annihilation removes dimensions	**Swap annihilation removes *distinguishability*, not dimensions**
SM visibility	Spectral (eigenvalue structure)	**Representation-theoretic (invisible to spectroscopy)**

The Standard Model gauge group SU(3)×SU(2)×U(1) is not a property of our universe. It is a theorem of mathematics. Its coupling constant α_GUT = $6/(25\pi^2)$ is not a measurement. It is a constant of ℂ.

Appendix A: Simulation Code and Reproducibility

All simulations were performed with NumPy (Python 3), seed=42, using normalized random vectors in ℂ^d. The swap operation symmetrizes paired blocks: ψ[block_i] ← (ψ[block_i] + ψ[block_j])/2, followed by renormalization. Effective rank is defined as the number of eigenvalues needed to capture 99% of Tr(G). Full code is available in the accompanying repository.

Appendix B: Why (3,2) and Not (2,3)?

The notation (n_S, n_T) = (3, 2) assigns the larger atom to space and the smaller to time. This is forced by chirality: only SU(2) has a pseudo-real fundamental representation ($2 \cong \bar{2}$ via ε_ab), which is the algebraic condition for chiral fermions in the weak sector. SU(3) does not have this property ($3 \ncong \bar{3}$), so it must be the spatial (vector-like confinement) sector.

Appendix C: Eigenvalue Convergence Data

C.1 Power Law Fit

The spectral gap $1 - \lambda_6/\lambda_5$ follows an approximate power law in N:

d	Fit	gap = α_GUT at N ≈

15	gap ≈ 0.786 × N^(−0.473)	N ≈ 1560
25	gap ≈ 0.453 × N^(−0.398)	N ≈ 1540

The exponent β ≈ 0.47 ≈ 1/2 is consistent with Marchenko-Pastur eigenvalue spacing (∝ 1/√N).

C.2 Convergence to Zero, Not to α_GUT

Extrapolating the power law:

N	gap (d=15)	gap/α_GUT
10^2	0.090	3.7×
10^3	0.028	1.2×
10^4	0.010	0.4×
10^{10}	0.000015	0.001×

The gap passes *through* α_GUT at N ≈ 1500 but does not stop there. This confirms that α_GUT is not a spectral quantity but a representation-theoretic/channel-counting quantity. The spectral gap is a finite-N effect that happens to coincide with α_GUT at a particular (physically relevant?) scale.

C.3 Open Question

The crossover N_cross ≈ 1500 (where gap = α_GUT) is remarkably consistent across different d values. Whether this N_cross has physical significance (e.g., as an effective vertex count per simplex neighborhood in our universe) is an open question.